Abhishek Subramanian

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Education

Virginia Tech — **Blacksburg, VA**
Master of Science in Computer Engineering - AI/ML and SWE concentration (**GPA 3.83 / 4.0**) — *Aug 2024 – May 2026*

College of Engineering, Guindy (Anna University) — **Chennai, India**
B.E. Electronics and Communications Engineering (**GPA 8.01 / 10.00**) — *Aug 2018 – Apr 2022*

Professional Experience

Quantiphi Inc. — **Boston, MA**
Machine Learning & Software Engineering Intern — *May 2025 – Present*

- Developed secure, **production-grade** AI systems for healthcare data intelligence and primary market research automation.
- Architected multi-agent pipelines using LangChain for search, summarization, supervision, and validation, integrating outputs from GPT-4, internal sources, authenticated sites, and external search results via SERP APIs.
- Built secure **Selenium** based extractors and applied an **LLM-as-a-judge** mechanism to detect hallucinations and ensure accuracy.
- Designed and implemented an **authN/authZ** framework for ~**60K** users using Microsoft Entra ID, **RBAC**, **PKCE**, **OAuth** propagation, and **MFA** to secure access to AI workflows.
- Developed middleware using **Streamlit**, **Vue**, **Next.js**, **FastAPI** for interacting with and inspecting agent outputs.
- Used **LangSmith** for tracing and evaluation to debug agent behavior and improve reliability.
- Deployed systems on **Azure** (App Services, App Containers, Blob Storage, **Azure OpenAI**), managing provisioning, model access, and operational scaling.
- Eliminated **30+** hours/week of manual work, reduced verification cycles by months, and enabled ~**$1.5M+** annual savings.

Quantiphi — **Bengaluru, India**
Full-Stack Software Developer — *Aug 2022 – May 2024*

- Led **backend development** for an ML-powered music synthesis platform, implementing authentication, music configuration, and server monitoring systems.
- **Architected data & API layer** with a secure **MySQL** DB (**5000+** concurrent users), **multithreaded processing**, and scalable **Flask APIs** (**CORS, WebSockets**) for real-time **frontend** communication and **Dockerized** services to reduce deployment time by **40%**.
- Constructed a full-stack **Generative AI-powered Digital Avatar chatbot** using **React.js** (**Redux** for state), **Material UI** components and **Express.js**, leveraging NVIDIA's Tokkio framework and integrating a **RAG** pipeline for real time contextual recommendations, boosting interaction metrics by **45%**.
- Engineered **gRPC API** calls to enable real-time processing and communication between NVIDIA Riva & Tokkio microservices and text based input systems, reducing cross-service latency by **~200ms** and ensuring seamless message synchronization.

Skills

Professional Certifications: Google Cloud Certified: Associate Cloud Engineer and Professional Cloud Developer
Programming Languages: Python, JavaScript, TypeScript, SQL, C++, C#, R, Java, HTML, CSS, Matlab.
Frameworks and Tools: Flask, Node.js, MySQL, PyTorch, TensorFlow, Git, Linux, NoSQL Firebase, React.js, Redux, Material UI, Bootstrap, Figma, Pygame, paho-mqtt, Pandas, FastAPI, PySpark, Android Studio, Hadoop, CUDA, LangChain, LangSmith.
Cloud and DevOps: Google Cloud Platform (GCP), NVIDIA Avatar Cloud Engine (ACE), Docker, Kubernetes, Jenkins, AWS.

Selected Projects

Explainable AI for Analyzing Transformer Models in NLP Tasks — *Jan 2025 – Present*

- Led a 3 person team in designing a **SHAP-informed Chain-of-Thought framework** with **shap.Explainer** on **zero-shot LLaMA 2** and **LLaMA 3** via Hugging Face Transformers, judged by Claude 3.7, achieving 31.84% win rate on LLaMA 3.
- Benchmarked SHAP-informed versus standard CoT explanations on the **e-SNLI** dataset using Partition SHAP with **prompt based conditioning**, obtaining a **22.35%** win rate for LLaMA 2 and demonstrating interpretability gains at scale.
- Formulated a scalable SHAP-based explanation pipeline in Python using Partition SHAP (with safety wrappers, limited evaluations, **greedy decoding prompts**) and Claude 3.7 Sonnet judge for 180 sample interpretability analysis.

Enhancing Emotional Well-Being through ML based Music Emotion Recognition — *Sep 2024 - Dec 2024*

- **Designed and fine-tuned CNN and RNN (LSTM) models** using MFCC-based features to capture temporal audio patterns, achieving **76.22%** **accuracy** on arousal-valence predictions using the DEAM dataset.
- **Validated emotion recognition performance** for mental health applications, demonstrating strong model generalization for therapeutic use cases.

Selected Publication

Unhealthy Liver Detection using CNN with IoT | IEEE ICSCSS 2023, Coimbatore, India — *Feb 2023 – Jun 2023*

- Presented a **CNN** inspired **CT** image analysis at the IEEE International Conference on Smart Computing and Smart Systems, achieving an **86.8%** detection rate and **30%** accuracy gain.
- Applied image processing pipelines **(augmentation, enhancement, restoration)** to diversify data and optimize performance.